

This document outlines the guidelines for developers to implement Career Aptitude Test score calculations and generate the Career Aptitude Test Report.

Career Aptitude Test Structure (Classes 6–12)

The Career Aptitude Test shall be conducted for students of **Classes 6, 7, 8, 9, 10, 11 and 12.**

Test Structure

- **Number of Sections:** 2
- **Section Name:**
 1. Interest & Personality Assessment
 2. Academic Proficiency Assessment
- **Number of Chapters:**
 1. Interest & Personality Assessment - 5
 2. Academic Proficiency Assessment - 5
- **Questions per Chapter:**
 1. Interest & Personality Assessment - 10
 2. Academic Proficiency Assessment - 10
- **Total Questions:** 100
- **Maximum Time Allowed:** 120 Minutes

Chapters

1. Interest & Personality Assessment

- 1) a. Medical Science & Healthcare and b. Science, Technology, Engineering & Mathematics
- 2) a. Commerce, Economics, Business & Management and b. Law, Politics, Social Work & History
- 3) a. Teaching, Coaching, Counselling & Education & b. Travel, Tourism, Hospitality, Aviation & Hotel Management
- 4) a. Administrative & Civil Services and b. Defense, Police, Sports & Yoga
- 5) a. Design, Branding, Fine Arts & Creativity and b. Performing Arts, Media, Journalism & Languages

2. Academic Proficiency Assessment

- 1) Science
- 2) Mathematics
- 3) Social Science
- 4) Language
- 5) Reasoning

Question Distribution

1. Interest & Personality Assessment

Chapter	Number of Questions
1) a. Medical Science & Healthcare and b. Science, Technology, Engineering & Mathematics	10
2) a. Commerce, Economics, Business & Management and b. Law, Politics, Social Work & History	10
3) a. Teaching, Coaching, Counselling & Education and b. Travel, Tourism, Hospitality, Aviation & Hotel Management	10
4) a. Administrative & Civil Services and b. Defense, Police, Sports & Yoga	10
5) a. Design, Branding, Fine Arts & Creativity and b. Performing Arts, Media, Journalism & Languages	10
Total	50

2. Academic Proficiency Assessment

Chapter	Number of Questions
1) Science	10
2) Mathematics	10
3) Social Science	10
4) Language	10
5) Reasoning	10
Total	50

Career Aptitude Test Report Generation Guidelines

Section 1: Interest & Personality Assessment

Area-wise Performance Calculation

The **Interest & Personality Assessment** measures a student's inclination towards different career disciplines based on their **interests, preferences, and personality traits**. Unlike a conventional examination, there are **no right or wrong answers** in this section. Each response contributes to one or more career disciplines depending on the option selected by the student.

The report shall display the student's aptitude across the following **10 Career Disciplines**:

1. Medical Science & Healthcare
2. Science, Technology, Engineering & Mathematics (STEM)
3. Commerce, Business & Management
4. Law, Politics & Social Work
5. Teaching, Coaching, Counselling & Education
6. Travel, Tourism, Hospitality, Aviation & Hotel Management
7. Administrative & Civil Services
8. Defence, Police, Sports & Yoga
9. Design, Branding, Fine Arts & Creativity
10. Performing Arts, Media, Journalism & Languages

Response Evaluation Logic

Each question is designed to represent **two different career disciplines**, where:

- ☐ **Option A** represents the **first career discipline**.
- ☐ **Option B** represents the **second career discipline**.
- ☐ **Option C (Both of the Above)** represents **both career disciplines**.
- ☐ **Option D (None of the Above)** does not represent either career discipline.

The scoring logic shall be implemented as follows:

Selected Option	Score Allocation
Option A	Award 1 point to the first career discipline only.
Option B	Award 1 point to the second career discipline only.

Selected Option	Score Allocation
Option C (Both of the Above)	Award 1 point each to both career disciplines.
Option D (None of the Above)	No points shall be awarded to either career discipline.

Example

Consider the following question:

Which of the following movies would you first prefer to watch?

- A.** Munna Bhai MBBS
- B.** 3 Idiots
- C.** Both of the Above
- D.** None of the Above

This question maps to the following career disciplines:

- ☐ **Option A** → Medical Science & Healthcare
- ☐ **Option B** → Science, Technology, Engineering & Mathematics (STEM)
- ☐ **Option C** → Medical Science & Healthcare **and** Science, Technology, Engineering & Mathematics (STEM)
- ☐ **Option D** → No career discipline

Therefore,

For Question Number 1 to 10:

Student Response	Score Awarded
Option A	Medical Science & Healthcare +1
Option B	Science, Technology, Engineering & Mathematics +1
Option C	Medical Science & Healthcare +1 and Science, Technology, Engineering & Mathematics +1
Option D	No score

For Question Number 11 to 20:

Student Response	Score Awarded
Option A	Commerce, Business & Management +1

Student Response	Score Awarded
Option B	Law, Politics & Social Work +1
Option C	Commerce, Business & Management +1 and Law, Politics & Social Work +1
Option D	No score

For Question Number 21 to 30:

Student Response	Score Awarded
Option A	Teaching, Coaching, Counselling & Education +1
Option B	Travel, Tourism, Hospitality, Aviation & Hotel Management +1
Option C	Teaching, Coaching, Counselling & Education +1 and Travel, Tourism, Hospitality, Aviation & Hotel Management +1
Option D	No score

For Question Number 31 to 40:

Student Response	Score Awarded
Option A	Administrative & Civil Services +1
Option B	Defence, Police, Sports & Yoga +1
Option C	Administrative & Civil Services +1 and Defence, Police, Sports & Yoga +1
Option D	No score

For Question Number 41 to 50:

Student Response	Score Awarded
Option A	Design, Branding, Fine Arts & Creativity +1
Option B	Performing Arts, Media, Journalism & Languages +1

Student Response	Score Awarded
Option C	Design, Branding, Fine Arts & Creativity +1 and Performing Arts, Media, Journalism & Languages +1
Option D	No score

Calculation of Aptitude Scores

The system shall maintain a separate score counter for each of the ten career disciplines.

Whenever a student selects an option, the corresponding career discipline(s) shall receive one point according to the response evaluation logic.

After all **50 questions** (25 Interest Test + 25 Personality Test) are completed, the application shall calculate the total score obtained in each career discipline.

The **Grand Total** shall be calculated as the sum of the scores obtained across all ten career disciplines.

Because **Option C** awards one point to both career disciplines represented in a question, the **Grand Total may exceed the total number of questions attempted**.

Aptitude Percentage Calculation

The report shall display:

- ☐ **Aptitude in Numbers**
- ☐ **Aptitude in Percentage (%)**

The percentage shall be calculated based on the student's **Grand Total**, which shall always be considered **100%**.

Formula

Grand Total = Sum of the scores of all 10 Career Disciplines

For each career discipline,

Aptitude (%) = (Career Discipline Score ÷ Grand Total) × 100

The calculated percentage shall be rounded to **two decimal places**.

Example

Suppose the student's scores are as follows:

Career Discipline	Aptitude in Numbers
Medical Science & Healthcare	6
STEM	4
Commerce, Business & Management	3
Law, Politics & Social Work	4
Teaching, Coaching, Counselling & Education	5
Travel, Tourism, Hospitality, Aviation & Hotel Management	4
Administrative & Civil Services	8
Defence, Police, Sports & Yoga	6
Design, Branding, Fine Arts & Creativity	4
Performing Arts, Media, Journalism & Languages	3
Grand Total	47

The percentages shall be calculated as:

$$\text{Aptitude (\%)} = (\text{Career Discipline Score} \div 47) \times 100$$

For example:

- ☐ Medical Science & Healthcare = $(6 \div 47) \times 100 = 12.77\%$
- ☐ STEM = $(4 \div 47) \times 100 = 8.51\%$
- ☐ Administrative & Civil Services = $(8 \div 47) \times 100 = 17.02\%$

The **Grand Total** shall always be displayed as:

- ☐ **Aptitude in Numbers:** 47
- ☐ **Aptitude in Percentage:** 100.00%

Developer Notes

- ☐ Each question shall map to exactly **two career disciplines**.
- ☐ Option **A** shall award **one point** to the first career discipline.
- ☐ Option **B** shall award **one point** to the second career discipline.
- ☐ Option **C** shall award **one point to both** career disciplines.
- ☐ Option **D** shall not award any points.

- ❑ Separate counters shall be maintained for all **10 career disciplines**.
- ❑ The **Grand Total** shall be calculated by adding the scores of all career disciplines.
- ❑ The **Grand Total** shall always be treated as **100%**, and the percentage for each career discipline shall be calculated relative to this value.
- ❑ All percentages shall be rounded to **two decimal places** before generating the Career Aptitude Test Report.

Section 2: Academic Proficiency Assessment

Area-wise Performance Calculation

The **Academic Proficiency Assessment** measures the student's academic performance across five subject areas based on the number of correct answers. Unlike the **Interest & Personality Assessment**, this section is evaluated using **correct and incorrect responses**, where each correct answer contributes to the student's proficiency in the corresponding academic subject.

The report shall display the student's performance in the following **five Areas of Academic Proficiency**:

1. Science
2. Mathematics
3. Social Science
4. Language
5. Reasoning

Each subject contains **10 questions**, resulting in a total of **50 questions** in the Academic Proficiency Assessment.

Accuracy Calculation

The report shall display the following information for each subject:

- ❑ **Accuracy in Numbers** (Number of Correct Answers)
- ❑ **Accuracy in Percentage (%)**

The **Accuracy in Numbers** shall represent the total number of correct answers obtained by the student in each subject.

Calculation of Subject-wise Accuracy

The application shall maintain a separate score counter for each of the five academic subjects.

For every correct answer:

- ☐ **1 mark** shall be awarded to the corresponding subject.
- ☐ Incorrect or unanswered questions shall receive **0 marks**.

After completion of the assessment, the system shall calculate:

Grand Total = Sum of Correct Answers in all Five Subjects

Accuracy Percentage Calculation

The report percentage shall be calculated **based on the student's total correct answers**, not on the total number of questions.

The **Grand Total** shall always be considered as **100%**, and the percentage shall be proportionally distributed among the five academic subjects.

Formula

Grand Total = Science + Mathematics + Social Science + Language + Reasoning

For each subject,

Accuracy (%) = (Subject Correct Answers ÷ Grand Total) × 100

The calculated percentage shall be rounded to **two decimal places**.

Example

Suppose a student obtains the following scores:

Academic Subject	Correct Answers
Science	8
Mathematics	9
Social Science	2
Language	6

Reasoning	8
Grand Total	33

The percentages shall be calculated as follows:

- ❑ **Science** = $(8 \div 33) \times 100 = \mathbf{24.24\%}$
- ❑ **Mathematics** = $(9 \div 33) \times 100 = \mathbf{27.27\%}$
- ❑ **Social Science** = $(2 \div 33) \times 100 = \mathbf{6.06\%}$
- ❑ **Language** = $(6 \div 33) \times 100 = \mathbf{18.18\%}$
- ❑ **Reasoning** = $(8 \div 33) \times 100 = \mathbf{24.24\%}$

The **Grand Total** shall always be displayed as:

- ❑ **Accuracy in Numbers:** 33
 - ❑ **Accuracy in Percentage:** 100.00%
-

Developer Notes

- ❑ Separate score counters shall be maintained for **Science, Mathematics, Social Science, Language, and Reasoning**.
- ❑ Only **correct answers** shall contribute to the subject score.
- ❑ Incorrect and unanswered questions shall not contribute to the score.
- ❑ The **Grand Total** shall be calculated as the sum of the correct answers obtained across all five subjects.
- ❑ The **Grand Total** shall always be treated as **100%**, and the percentage for each subject shall be calculated relative to this value.
- ❑ All percentage values shall be rounded to **two decimal places** before generating the Academic Proficiency Assessment Report.
- ❑ The sum of all subject-wise percentages shall always equal **100.00%** (allowing only minor rounding differences due to decimal rounding).

Report Template



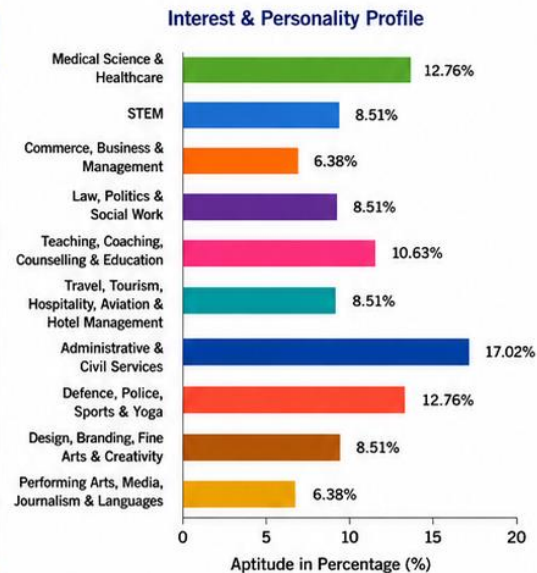
CAREER APTITUDE TEST REPORT

Student Name : Adiraj Sunil Patil
Class : 9

SECTION 1: INTEREST & PERSONALITY ASSESSMENT

This section evaluates your interests and personality traits across different career domains.

Career Interest Areas	Aptitude in Numbers	Aptitude in Percentage (%)
1. Medical Science & Healthcare	6	12.76%
2. Science, Technology, Engineering & Mathematics (STEM)	4	8.51%
3. Commerce, Business & Management	3	6.38%
4. Law, Politics & Social Work	4	8.51%
5. Teaching, Coaching, Counselling & Education	5	10.63%
6. Travel, Tourism, Hospitality, Aviation & Hotel Management	4	8.51%
7. Administrative & Civil Services	8	17.02%
8. Defence, Police, Sports & Yoga	6	12.76%
9. Design, Branding, Fine Arts & Creativity	4	8.51%
10. Performing Arts, Media, Journalism & Languages	3	6.38%
GRAND TOTAL	47	100.00%

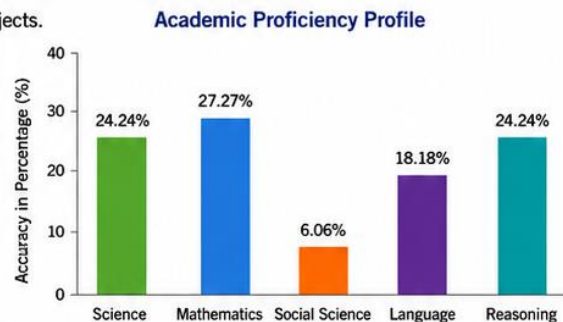


Insights: Your strongest inclination is towards Administrative & Civil Services, followed by Defence, Police, Sports & Yoga and Medical Science & Healthcare.

SECTION 2: ACADEMIC PROFICIENCY ASSESSMENT

This section evaluates your academic performance across core subjects.

Academic Subjects	Correct Answers (in Numbers)	Accuracy in Percentage (%)
1. Science	8	24.24%
2. Mathematics	9	27.27%
3. Social Science	2	6.06%
4. Language	6	18.18%
5. Reasoning	8	24.24%
GRAND TOTAL	33	100.00%



Insights: You have performed the best in Mathematics, followed by Science and Reasoning. Keep strengthening Social Science to achieve a balanced overall performance.

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